Force Admittance Control of An Underactuated Gripper with Full-State Feedback

Chunpeng Wang[1], David Nguyen[2], Zhi Ern Teoh[1], Ciaran ONeill[1], Lael Odhner[1], Peter Whitney[1,3,4], and Matthew A. Estrada[1,4]

***Abstract*— We present admittance control and fingertip contact detection with a linkage gripper remotely driven by a pneumatic rolling diaphragm actuator. The gripper is driven by underactuated mechanisms sensorized by joint encoders in order to fully determine the gripper state. We present the modelling of the linkage and fluidic transmission, validate its ability to regulate pinch force via admittance control within an RMS error well under 0.5 Newtons, and show the ability to detect contact at targeted locations on the linkage. In addition, we demonstrate simple grasping behaviors: blindly searching for an unobstructed object and detecting object loss. Our results show that an integrative approach of instrumenting underactuated gripper mechanisms can result in a lightweight gripper that is not only mechanically adaptive but sensitive enough to react to contact events without distal sensors or vision.**

I. INTRODUCTION

Realizing reactive, contact-sensitive grippers has been a central goal of robotic manipulation since its inception [1]. Robotic gripper design is one consideration on how to gracefully interface with the external environment. Design often lies on a spectrum between rigid grippers amenable to more classical modelling and control methods, and, in contrast, highly deformable grippers that are mechanically adaptive albeit more difficult to model.

Mechanically adapting to the environment via underactuation or soft structures is a commonly employed strategy. Compared to fully actuated systems, underactuated systems are often more cost-effective and simpler in design. A notable example is in the design of robotic hands [2]–[5]. By carefully selecting appropriate passive stiffness and implementing hard stops, the functionality of the fingertips can adapt to different phases of the manipulation process, allowing for more sophisticated and dexterous operations [6]. Further still, soft or particle-jamming grippers take this principle to a continuum [7].

On the opposite extreme, many industrial applications rely on parallel jaw grippers [8] or suction, in large part due to their simplicity. Though less articulated, they are functional and often simple enough to model with discrete states (e.g. open/closed). In order to extend functionality, there is a tendency to consider sensors after mechanical design, or completely separate. This approach offers modularity but can also limit the ability to cohesively integrate solutions. In the examples of underactuated fingers in [9], [10], admittance control is utilized to achieve the desired grasping force also based on tactile sensor signals.

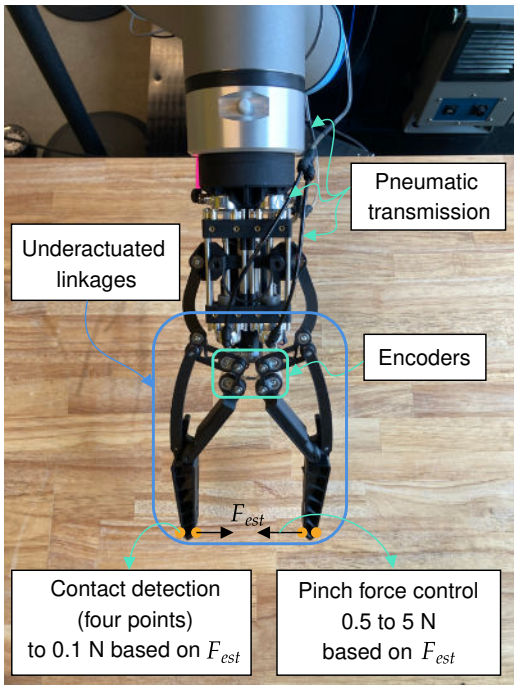


Fig. 1: The pneumatic underactuated linkage gripper. Linkage kinematics and joint encoders provide full state feedback of the underactuated system thus allowing force estimation at the fingertips without distal sensors.

Integrated efforts can merge gripper or transmission mechanics and sensing into the same structures. The exploratory hand in [11] demonstrates an integrative design of a low-impedance end-effector, and that tactile sensors further aid estimating the object location. The Direct Drive Hand in [12] provides high-fidelity transmission of force and motion signals by the linkage design and the use of direct drive motors. Previous work in [13] shows contact force can be estimated utilizing a multi-DOF lumped inertia-spring model without external tactile sensors, extending the Series Elastic Actuator (SEA) model [14] to hydrostatic transmission systems for estimating frequencies beyond quasi-static range.

In this work, we present force estimation and control of a rolling-diaphragm based, remote direct drive gripper shown in Fig. 1. We demonstrate an admittance controller

[1]RAI Institute, Cambridge, MA 02142, USA
[2]Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139, USA
[3]Department of Mechanical and Industrial Engineering, Northeastern University, Boston, MA 02115, USA
[4]corresponding author, j.whitney@northeastern.edu, mestrada@rai-inst.com

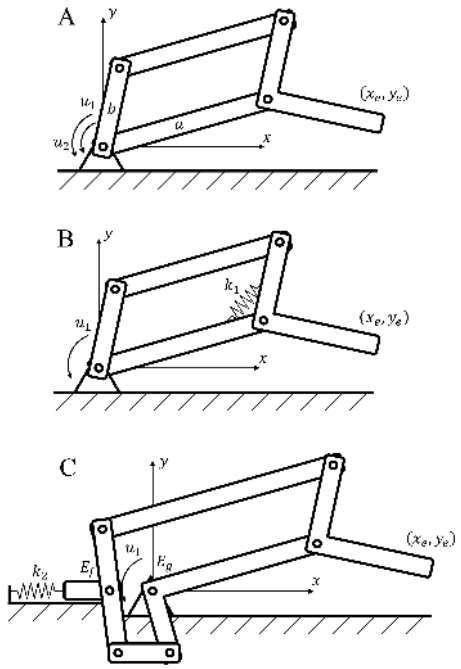


Fig. 2: 2D Planar Finger Linkages. Linkage A is fully actuated with u_1 and u_2 controlling links a and b respectively. Linkage B is underactuated with k_1 replacing u_2. Linkage C offers a hybrid mechanism that moves the underactuation spring to k_2.

to command desired finger pinch force and desired finger location. The virtual admittance model is selected based on passivity to ensure the closed-loop system stability. We validate that the pinch force is controllable and the estimated force can be utilized to detect external contact. Two examples are shown to demonstrate the gripper capabilities in potential applications.

II. MECHANICAL LINKAGE DESIGN

We introduce a new linkage mechanism implemented in a lightweight planar gripper shown in Fig. 1. The gripper was combined with remotized rolling diaphragm actuation, seen in Fig. 3, to offer high force output and speed.

Traditionally, robot grippers often use a common four or five-bar linkage mechanism for 2D planar position control of the fingertip. The design of the linkage allows for full control of the finger stage with grounded actuators u_1 and u_2 controlling links a and b respectively in Fig. 2 A. Often the actuator u_2 is replaced with a spring k_1 and hard stop making the finger underactuated [15] as seen in Fig. 2 B. Approach A offers superior measurement and control of the finger while B is simpler and has less reflected inertia.

Instead of embedding a spring within the gripper linkage, we implement a hybrid option seven-bar mechanism in Fig. 2 C with a grounded spring. The underactuation is now enabled by a prismatic joint and grounded spring removing the mass and complexity from the distal links. The additional joints also allow for encoders E_f and E_g to measure the state of the finger for proprioceptive contact detection.

III. LINKAGE KINEMATICS AND JACOBIAN

We derive the kinematics of the linkage gripper, which ultimately allows to estimate position and force at the finger tips from proximal encoder and pneumatic air hose force signals. The kinematics of the system computes the position of the finger's distal link and estimating the forces $F_{A,x}$ and $F_{T,x}$ from the driving forces F_{air} and F_{spring}. In order to derive these relationships, the entire linkage is broken down into three components (colors refer to Fig. 3):

- Finger proximal and distal links linkage (cyan links): this linkage provides the primary calculation of the finger distal link motion.
- Double rocker-like linkage (red links): this linkage interacts with the finger proximal and distal links linkage to form a closed kinematic chain, which is crucial for maintaining the 2-DOF system configuration.
- Slider-rocker-like linkage (purple links): this linkage provides a driving mechanism from linear actuator to drive the seven-bar linkage.

By decomposing the finger linkage system into three subsystems—namely, the double rocker-like linkage, the slider-rocker-like linkage, and the five-bar-like finger proximal and distal links linkage—we can compute the fingertip gap using the two encoder readings. Furthermore, these kinematic equations are instrumental in constructing the Jacobian matrix, which is essential for estimating the pinch force exerted on the finger's distal links.

The force input for each finger comes from two sources: the driving force F_{air} from the air hose and the spring force F_{spring} acting on the slide in the y-direction. To estimate the pinch force on the distal link of the finger, we need to calculate the Jacobian matrix J_a defined by:

$$\begin{bmatrix} \dot{y}_{E_f} \\ \dot{y}_P \end{bmatrix} = J_a \begin{bmatrix} \dot{x}_T \\ \dot{x}_A \end{bmatrix}. \tag{1}$$

The estimated forces $\hat{F}_{T,x}$ and $\hat{F}_{A,x}$ on the distal link are then computed using the transpose of J_a as:

$$\begin{bmatrix} \hat{F}_{T,x} \\ \hat{F}_{A,x} \end{bmatrix} = J_a^T \begin{bmatrix} \hat{F}_{spring} \\ \hat{F}_{air} \end{bmatrix}. \tag{2}$$

Here, the estimated spring force $\hat{F}_{spring}$ is calculated from the product of known spring stiffness and the spring deflection. The estimated force from the air hose $\hat{F}_{air}$ also exhibits a linear hookean behavior, which is calculated from the product of the identified air hose stiffness and the relative position error between motor and gripper driving piston position. Targeted experiments were conducted to identify these values.

The computation of the Jacobian matrix J_a can be separated into two parts as follows:

$$\begin{bmatrix} \dot{y}_{E_f} \\ \dot{y}_P \end{bmatrix} = J_1 \begin{bmatrix} \dot{\theta}_g \\ \dot{\theta}_f \end{bmatrix}, \begin{bmatrix} \dot{x}_T \\ \dot{x}_A \end{bmatrix} = J_2 \begin{bmatrix} \dot{\theta}_g \\ \dot{\theta}_f \end{bmatrix} \Rightarrow J_a = J_1 J_2^{-1}. \tag{3}$$

Here, J_1 defines the Jacobian matrix relating the joint velocities $\dot{\theta}_g$ and $\dot{\theta}_f$ to the velocities $\dot{y}_{E_f}$ and $\dot{y}_P$, and J_2 defines the Jacobian matrix relating the joint velocities $\dot{\theta}_g$ and

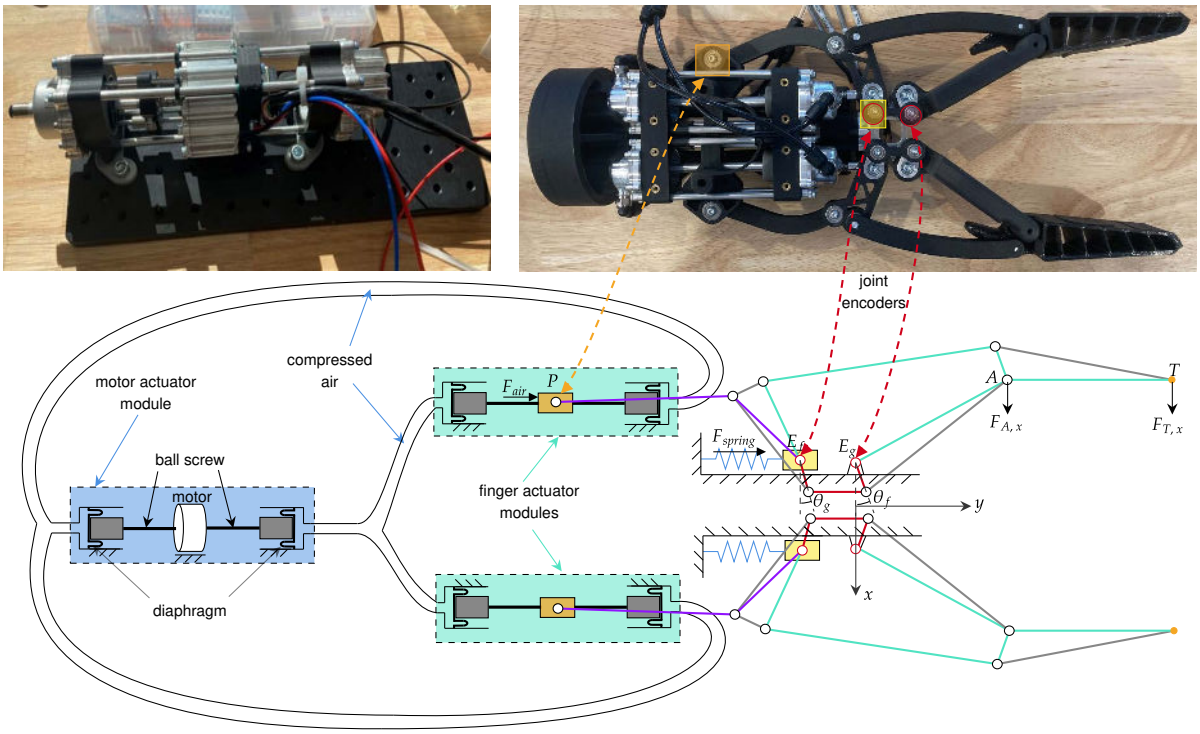


Fig. 3: Underactutated gripper hardware and its schematic. One motor module is used to drive to two finger linkages. The joint angles θ_g and θ_f are obtained from encoder readings. The driving forces F_{air} and F_{spring} are estimated from the air hose and the coil spring, respectively. $F_{A,x}$ and $F_{T,x}$ are the finger distal link forces in the gripper open / close direction.

$\dot{\theta}_f$ to the velocities $\dot{x}_T$ and $\dot{x}_A$. J_1 and J_2 can be computed from forward kinematics by taking the partial derivative with respect to θ_g and θ_f. The forward kinematics and Jacobian equations are written as a C++ function, with input of linkage configuration, encoder readings, and estimated force.

The precision of estimated force $\hat{\mathbf{F}} = [\hat{F}_{T,x}, \hat{F}_{A,x}]^T$ is dependent on the encoder input $\mathbf{\Theta} = [\theta_g, \theta_f]^T$ as:

$$\begin{aligned}\frac{d\hat{\mathbf{F}}}{d\mathbf{\Theta}} &= J_a^T \frac{d\begin{bmatrix}\hat{F}_{spring}\\ \hat{F}_{air}\end{bmatrix}}{d\mathbf{\Theta}} = J_a^T \begin{bmatrix}\hat{k}_{spring} & 0\\ 0 & \hat{k}_{air}\end{bmatrix} \frac{d\begin{bmatrix}-y_{E_f}\\ -y_P\end{bmatrix}}{d\mathbf{\Theta}} \\ &= -(J_2^{-1})^T J_1^T \begin{bmatrix}\hat{k}_{spring} & 0\\ 0 & \hat{k}_{air}\end{bmatrix} J_1.\end{aligned} \tag{4}$$

Eq. (4) tells us the force changes with respect to encoder input changes, and can be used to roughly calculated the threshold for force detection. However, J_1 and J_2 are dependent on θ_g and θ_f, meaning that the sensitivity of force is not a constant.

IV. Force Admittance Control

In contrast to a position-based control interface, where the gripper's movements are predetermined based on the estimated size and shape of the object, a force control interface offers a more adaptive approach to object manipulation. By focusing solely on force feedback, this method simplifies the control strategy, reduces the computational load associated with object estimation.

The model-based force control in [16] demonstrates a admittance control to reduce impedance to a pure inertia. Instead of a virtual damper connection between the virtual model and the physical model, a parallel virtual spring and damper connection is preferred to minimize the error between the virtual model and the actual position, with the control law shown as:

$$F_c(s) = (K_v + B_v s)\left(\frac{F_d(s) - F_e(s)}{M_v s^2} - X(s)\right), \tag{5}$$

where F_c is the control force, F_d and F_e are the desired force and external force respectively, M_v is the virtual model inertia, B_v and K_v are the virtual parallel damping and stiffness, X is the actual physical position. The closed-loop admittance with the equation of motion $F_c(s) + F_d(s) - F_e(s) = (ms^2 + bs)X(s)$ can be calculated as:

$$Y(s) = \frac{sX(s)}{F_d(s) - F_e(s)} = \frac{1}{M_v}\frac{M_v s^2 + B_v s + K_v}{ms^3 + (b + B_v)s^2 + K_v s}. \tag{6}$$

According to the 1-port LTI passivity criterion in [17], the parameters require the following condition:

$$M_v \geq \frac{B_v}{b + B_v} m, B_v \geq 0, K_v = 0. \tag{7}$$

One viable approach to maintaining a positive K_v is by introducing additional damping B_a into the virtual model. The adjusted control law can be expressed as:

$$F_c(s) = (K_v + B_v s)\left(\frac{F_d(s) - F_e(s)}{M_v s^2 + B_a s} - X(s)\right). \tag{8}$$

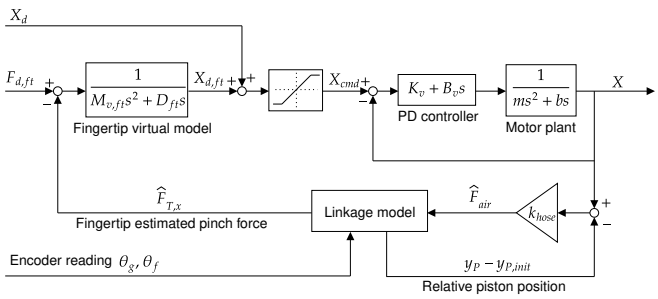


Fig. 4: Underactuated gripper fingertip pinch force admittance controller.

The admittance with the updated controller is shown as:

$$\begin{aligned} Y(s) =& \frac{M_v s^2 + (B_v + B_a)s + K_v}{D(s)}, \\ D(s) =& M_v m s^3 + (M_v(b + B_v) + mB_a)s^2 \\ &+ (B_a(b + B_v) + M_v K_v)s + B_a K_v. \end{aligned} \tag{9}$$

According the same 1-port LTI passivity criterion, M_v and B_v maintain the same range limit from Eq. (7), and we can choose a positive K_v. The added damping B_a needs to satisfy the following condition as:

$$\begin{cases} (b + B_v)B_a^2 + (B_v(b + B_v) - mK_v)B_a - M_v b K_v \geq 0 \\ B_a \geq 0 \end{cases} \tag{10}$$

Given that the parameters are all positive, there always exists a positive solution of B_a that satisfies the quadratic function in Eq. (10) to be positive. Consequently, the passivity of the closed-loop system is preserved.

This result demonstrates that by introducing damping into the virtual model, we can mitigate the potential position drift when $K_v = 0$, allowing for the use of a lower-level position PD controller. While this approach may involve a trade-off in performance due to the added damping, it provides a practical solution for maintaining stability and avoiding drift.

The fingertip pinch force controller is shown in Fig. 4 A. We preset the largest stable value of K_v and B_v in the micro-controller interface to promote high bandwidth and responsiveness. Then, we choose the minimum value of fingertip's virtual inertia $M_{v,ft}$ and virtual damping $D_{v,ft}$ which stabilizes the system. The fingertip desired position offset from the error between desired contact force and the estimated force is defined as:

$$X_{d,ft}(s) = (M_{v,ft}s^2 + D_{ft}s)^{-1}(F_{d,ft}(s) - \hat{F}_{T,x}(s)). \tag{11}$$

The resulting position offset $X_{d,ft}$ is added back into the motor position controller as:

$$X_{cmd}(s) = X_d(s) + X_{d,ft}, \tag{12}$$

where X_{cmd} is the commanded motor position, and X_d is the feedforward desired motor position without force control. The commanded motor position X_{cmd} is then used in the preset PD controller as:

$$F_c(s) = (K_v + B_v s)(X_{cmd}(s) - X(s)). \tag{13}$$

This setup enables the selective activation of force control. For instance, when the gripper is closing to grasp an object, force control can be activated to ensure the object is securely held. Conversely, when the gripper is opening, force control can be deactivated to prevent position drift caused by force estimation errors. This flexibility allows the system to maintain stability and accuracy in various operational scenarios.

V. EXPERIMENTAL RESULTS

We present our experimental hardware setup, validation of fingertip force estimation, and several basic demonstrations of reactive behaviors it enables. Videos of demonstrations can be found in the supplementary materials.

A. Hardware Setup

The underactuated gripper in Fig. 3, consists of a palm with two fingers, each constructed as a nine-bar linkage system (including the driving links). Each linkage system features an actuated joint powered by compressed air through a diaphragm-based linear actuator, along with an unactuated joint slider equipped with a passive spring with a stiffness of 5067 N/m. To monitor the link states of each linkage system, two incremental encoders (USDIGITAL E16-4096-079-S-D-D-B) are employed. These encoders are connected to an MCC USB-QUAD08 Data Acquisition device to capture joint position data.

The gripper's diaphragm linear actuator is driven by compressed air, with two air hoses connected to a motor actuator module, forming a double actuation system. The hose is pressurized at 100 PSI and the stiffness of air hose is estimated as 12000 N/m. The equivalent stiffness of air hose is estimated by the ratio of motor torque versus the position difference between motor and gripper side piston, when commanding a sinusoid motor position from 0 to 10 Hz. The motor actuator module is a motorized diaphragm-based linear actuator, where a ball screw converts the motor's rotational motion into linear movement to push the piston in two directions. The piston is driven by an Akribis ADR50-T-14-P motor, paired with a Quantum Device LP12-20000-8-A-C-E-C-A encoder, and controlled via a Copley Nano drive. Communication between the local Linux machine and the motor drive is through EtherCAT at a rate of 1 kHz. A ROS2-based interface shown in Fig. 4 B, built using the motor drive's C++ library, is employed to receive and send information for high-level command execution also at 1 kHz.

B. Underactuated Gripper Fingertip Force Estimation

We validated fingertip pinch force using a load cell (Pinch Dynamometer by BIOMETRICS LTD, with a rated load capacity of 0 to 22.68 kg) positioned between the two fingers of the gripper. The force measured by the load cell was considered to be the true force $F_{T,x}$. As shown in Fig. 5, the estimated force (green curve) closely tracks the target force (blue curve). The RMS error between the measured force and the estimated force remains under 0.5 N. The experiment shows the force estimation RMS error is smaller with larger fingertip gap. This is mainly due to the nature of

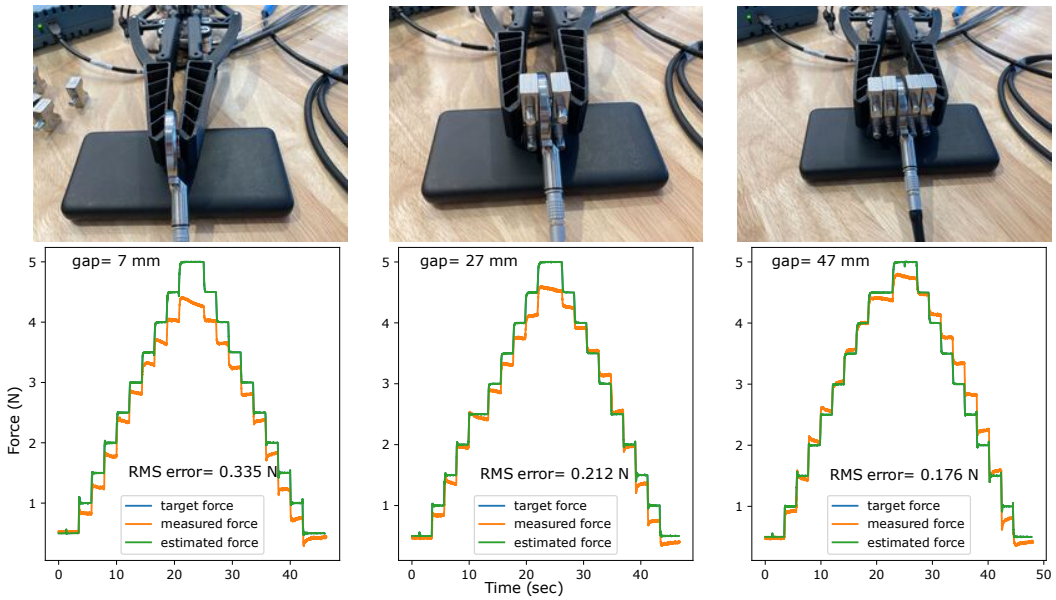


Fig. 5: Experiment evaluating estimated pinch force at fingertip. The commanded pinch force starts at 0.5 N, increases to 5.0 then reduces to 0.5 N, in increments of 0.5 N lasting for 2 seconds each. The fingertip gap is varied from 7 to 47 mm (7 mm from the load cell and 20 mm from each steel block) RMS error is well below 0.4 N for all of the grasp widths.

compressible air and the actuator piston of gripper reaching the hard stop. With fingertip gap of 47 mm, the actuator piston approaches the hard stop with 4.5 N pinch force. In the plot of 7 mm gap width, another contribution of error is due to the sensor geometry. The direction of measured force is perpendicular to the left fingertip, and the measured force is only partial of the true pinch force, which is the reason why measured error is larger even before reaching the hard stop.

C. Demo: underactuated gripper fingertip contact detection

The estimated fingertip force $\hat{F}_{T,x}$ serves as a reliable indicator for determining whether the fingertip is in contact. We programmed a primitive to detect light contact with an external object while the end-effector frame was in motion, and reverse direction if so. A contact force threshold of 0.1 N was set based on the estimated fingertip force. The choice of force threshold is constrained by the gripper frame acceleration and related joint deflection due to the inertial load of fingertip. The finger contact events match the demonstration from the supplementary video.

D. Demo: grasping via contact events

Force estimation is also useful for contact-based grasping, where the estimated force indicates if an object has been touched or is positioned between the two fingers of the gripper. In our contact-event-based grasping test, a bottle can was placed on a table as the target object. The arm searches for the object by moving along its fingertip's local x-axis. A three-step process is applied to determine when to close the fingers for grasping, as illustrated Fig. 7.

This simple contact event based heuristic shows the base capability for more generalized routines using discrete contact events in combination with surrounding environmental context, such as grasping with POMDPs [18].

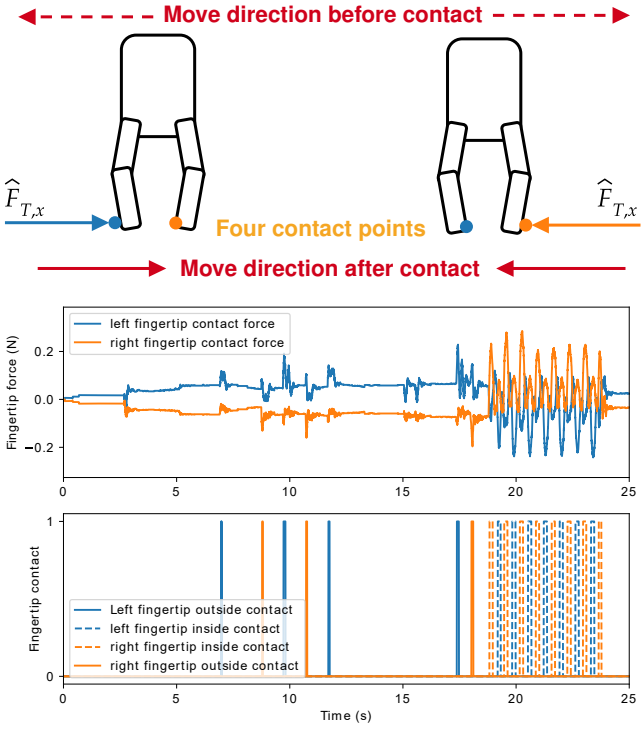


Fig. 6: Underactuated gripper fingertip contact experiment with arm reflex indicator based on the force estimation.

E. Demo: object loss detection

Object loss detection is shown to be quickly discerned from fingertip grasp width. In the experiment, the gripper closes its fingers with a target 1.2 N pinch force to pick up a baseball (approximately 75 mm diameter) and the fingertip

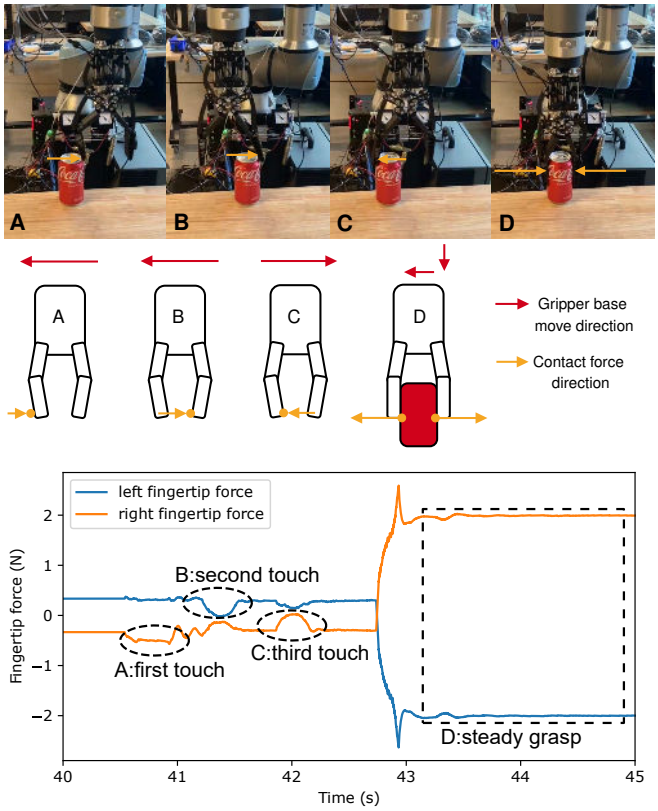


Fig. 7: Contact-event-based grasp experiment. (A) First touch with first fingertip outside surface; (B) second touch with the second fingerip inner surface; (C) third touch with the first fingertip inner surface; (D) steady state grasp.

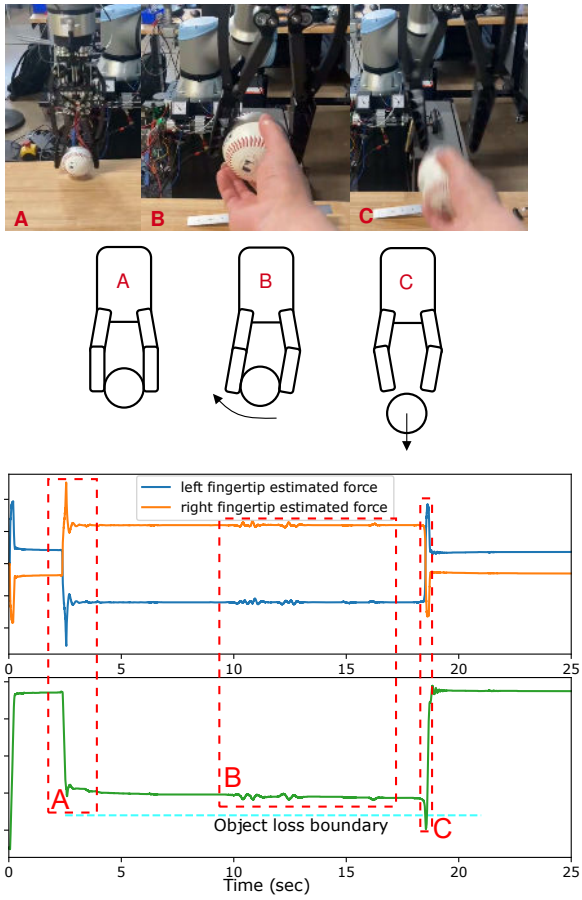


Fig. 8: Object loss experiment. (A) Grasp the base ball. (B) The operator wiggles base ball without triggering loss signal. (C) The operator removes the base ball and trigger loss detection signal based on fingertip gap.

gap acts as an indicator to determine if the ball is lost in Fig. 8. If the ball is lost, the gripper controller recognizes the event and commands an open position. The results show the disturbance of gripper fingertip contact (i.e. disturbing the object in-hand while the grasp is maintained) is not triggered as object loss if the object loss threshold is carefully chosen (a threshold of 0.01 m for the fingertip gap change is set in this experiment).

VI. CONCLUSIONS

In this work, we present a method for fingertip-level force estimation and control of an underactuated gripper. By utilizing select signals, such as the gripper's driving force and the joint positions of the finger linkage, we demonstrate that it is possible to achieve several practical grasping functionalities without relying on tactile sensors located at the gripper's fingertips, nor vision.

Our experiments verified that both the fingertip gap and the pinch force can be estimated within an RMS error of 0.5 Newton. The gripper is capable of detecting contact in the direction of fingertip's local x-axis. This contact indicator can be further utilized for blind searching within the gripper plane. In the final experiment on object loss detection, the system is able to identify if an object has been lost from the gripper immediately due to any disturbance, based on the steady fingertip gap change.

Controlling around instrumented, passive degrees of freedom indeed comes with some drawbacks, such as reducing the bandwidth with which the gripper can be driven or not being able to drive some joints at all. The advantage is that grasping and general operation benefit from improved stability and predictability when transitioning across contact events, as shown in our demonstrations. Especially when intended usage is predictable (e.g. contact near fingertips in this work), a paradigm of thoughtful design and sensing of underactuated mechanisms may well attain a base level of functionality before further perception is added by striking a practical balance that is mechanically adaptive yet provides full-state feedback.

Future work will focus on addressing potential uncertainties in detection and actuation. The current methods are deterministic and rely on clean, reliable signals at the time of decision-making; any signal noise or disturbances can lead to failure in task execution. Developing more robust algorithms and incorporating strategies to handle noisy or uncertain environments will be essential for improving the reliability and adaptability of the system in real-world applications.

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